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Th1. Hermiticity

Th1–Th6: hermiticity, self-adjointness, certificates, the spectral Li formula, block reduction, the Guinand–Weil trace formula

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Th1. Hermiticity and reality of the spectrum

Theorem Th1 (V4).

For any choice of vortices, W, α and boundaries the operator H_L satisfies H_L = H_L†; hence spec(H_L) ⊂ ℝ. At α = ½ and W = 0 the spectrum is symmetric: λ ∈ spec ⇒ −λ ∈ spec.

> ***Proof.*** *Every bond is inserted as the pair H[i,j] = −t·e^{iφ}, H[j,i] = −t·e^{−iφ} = conj(H[i,j]) — hermiticity by construction; the diagonal is real (vortex potential, disorder, staggered mass). By the spectral theorem for Hermitian matrices the whole spectrum is real. Symmetry at W = 0: the lattice is bipartite in the parity ix+iy; in that basis H = [[0, B], [B†, 0]]; if H(λ, 0) = λ·(u, v) then H(−v, u)·(−1) = −λ·(u, v) — the spectrum is symmetric about zero. Numerical certificate: ‖H − H†‖ = 0.0 exactly; the pairing λ ↔ −λ reproduced to 8.2e-15; reality residuals 8.8e-15 (the eigvalsh machine floor).*

Th1.1 Certificate and boundaries

The certificate is reproducible: test_th1_passport.py checks hermiticity on three protocol gates (Nv = 16/64/256, seed 96) and demands ‖H − H†‖ ≤ 1e-12; the fact is exactly 0.0, since the construction uses no approximations — conjugate pairs are written simultaneously. Importantly, hermiticity survives ALL the C5 handles — twists, anisotropy, δα, staggered mass, profile, signs: each handle either modifies the bond magnitude (preserving the phase pair) or the diagonal (staying real). This makes Th1 an architectural property, not a property of a single configuration.

Boundaries: Th1 says nothing about the correspondence of the spectrum to the ζ zeros — that is the H2/H5 question, where the series' verdicts are: the level identity holds at the zeros' level (V6/V16 — YES via the D_nn bridge, v2.0.7) with the bit-level branch open (V16, z = 2.49 < 6), and the specific design claims refuted (V1: the blind coding is invisible; V3: σ = ½ is not optimal in D_KS). The pairing λ ↔ −λ breaks already at W > 0 (the vortex potential is asymmetric) and is not restored by vortices.